Supplementary Information: The resolution paradox in skill-demand forecasting: Mosaic cross-scale learning across regional and occupational labour markets

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This Supplementary Information provides the forecasting protocol, implementation details, additional checks and robustness analyses supporting the main results.

S1. Data provenance and aggregation audit

The analysis uses the public processed Job-SDF demand tables. The L1 and L2 bottom lattices are indexed by skill, region, occupation and month, and direct summation reproduces the market, regional and occupational margins used by the analysis. The source repository snapshot and exact input files are fixed by commit, file size and SHA-256 digest in Online Resource 2.

The released processed tables contain 2,335 contiguous skill IDs (0-2334). The Job-SDF main text reports 2,324 standardised skills, while Appendix F.2 of the same paper reports 2,335 skills over 36 months. Table S1 applies the current strong-local primary specification after excluding IDs 2324-2334 and compares the estimates with the 2,335-ID analysis.

Table S1 Skill-count sensitivity after excluding the 11 highest-numbered IDs (2,324 IDs retained) under the primary strong-local specification.

Level	Skills	Strong local AP	Full Mosaic AP	Gain
L1	2,335	0.320	0.371	0.051
L1	2,324	0.319	0.372	0.053
L2	2,335	0.283	0.338	0.055
L2	2,324	0.283	0.338	0.055

S2. Forecasting protocol

The forecasting protocol uses a 12-month recent-history window and a three-month horizon. Three non-overlapping 2022 target blocks are used to fit the primary GBM models. October-December 2022 is excluded from primary GBM fitting and serves as the validation block for the secondary neural comparison. The primary strong-local and full-Mosaic models are each fitted once and then held fixed across the four quarterly 2023 forecast origins; only predictor histories advance as additional months become observed. Each forecast uses only information observed before its forecast origin.

Recent local event observability is the number of positive months in the preceding 12-month focal history. A cell is dormant when the final pre-origin month is zero; the rare-dormant subset requires at most two active months. The target is positive when at least one of the next three months contains demand. Quarterly estimates remain positive at both resolutions (Table S2).

Table S2 Rare-dormant strong-local and full-Mosaic AP by 2023 forecast block.

Level	Forecast block	Strong local AP	Full Mosaic AP	Gain
L1	2023-01 to 2023-03	0.303	0.358	0.055
L1	2023-04 to 2023-06	0.320	0.374	0.054
L1	2023-07 to 2023-09	0.342	0.396	0.054
L1	2023-10 to 2023-12	0.315	0.360	0.044
L2	2023-01 to 2023-03	0.264	0.320	0.057
L2	2023-04 to 2023-06	0.282	0.344	0.061
L2	2023-07 to 2023-09	0.296	0.353	0.057
L2	2023-10 to 2023-12	0.288	0.335	0.047

S3. Mosaic feature construction and learner settings

The primary local representation contains the log-transformed demand counts for the 12 focal months, recent active-month count, log-transformed final local count, cumulative pre-origin active months, the corresponding active-month ratio, log-transformed months since the most recent positive observation, and an indicator for activity before the current 12-month window. Mosaic retains this complete focal representation and adds three disjoint 12-month histories: regional, occupational and residual-market context. Inclusion-exclusion removes overlap among the three external histories.

Mosaic-GBM uses 45 trees, 25 leaves, learning rate 0.07, row subsampling 0.8 with bagging enabled, feature subsampling 0.8, L2 regularisation 1.0, balanced class weights and random seed 2026. The released primary GBM pipeline uses this fixed specification directly and does not perform a separate validation-based hyperparameter search. Strong-local and full-Mosaic models are each fitted once on the three 2022 training blocks and held fixed across all 2023 evaluation origins. The local reference uses the same learner configuration and focal information; Mosaic additionally receives the cross-scale histories.

The bin-level matched effect and its paired skill-cluster intervals are reported in Table S3. The rare-dormant aggregate contrast is isolated in Table S4.

Table S3 Strong matched recent-local-event-observability gradient with 1,000-replicate paired skill-cluster intervals.

Level	A	N (000s)	Gain	95% CI
L1	0	503.5	0.076	[0.073, 0.079]
L1	1	69.5	0.042	[0.037, 0.047]
L1	2	35.9	0.029	[0.024, 0.035]
L1	3-5	51.5	0.019	[0.016, 0.022]
L1	6-11	30.8	0.003	[0.002, 0.004]
L2	0	2493.1	0.074	[0.071, 0.076]
L2	1	202.9	0.040	[0.038, 0.043]
L2	2	95.8	0.034	[0.031, 0.038]
L2	3-5	125.8	0.017	[0.015, 0.019]
L2	6-11	66.5	0.001	[-0.000, 0.001]

Table S4 Paired skill-cluster intervals for the rare-dormant full-minus-strong-local AP gain.

Level	Observed AP gain	2.5%	97.5%	Replicates
L1	0.051	0.049	0.053	1,000
L2	0.055	0.054	0.056	1,000

S4. Statistical uncertainty and prevalence sensitivity

The primary observability-gradient and rare-dormant intervals use 1,000 paired skill-cluster resamples. Each evaluated subset resamples only the skill clusters represented in that subset, keeping all regions, occupations and forecast origins for a sampled skill together. L1 context-attribution intervals use 500 resamples with the same cluster design. These intervals are conditional on the fitted models and the observed 2023 evaluation blocks.

Average precision changes with positive-class prevalence. Table S5 reports the prevalence-normalised score (AP - prevalence) / (1 - prevalence) under the same strong-local specification as the primary analysis. The gain remains concentrated in low-observability bins.

Table S5 Prevalence-normalised recent-event-observability gradient under the primary strong-local specification.

Level	A	Prev.	Local AP	Full AP	Raw gain	Norm. gain
L1	0	0.041	0.123	0.199	0.076	0.079
L1	1	0.226	0.352	0.394	0.042	0.055
L1	2	0.359	0.491	0.521	0.029	0.046
L1	3-5	0.560	0.699	0.718	0.019	0.043
L1	6-11	0.847	0.934	0.936	0.003	0.017
L2	0	0.024	0.100	0.174	0.074	0.076
L2	1	0.203	0.327	0.368	0.040	0.051
L2	2	0.337	0.467	0.501	0.034	0.052
L2	3-5	0.535	0.680	0.697	0.017	0.037
L2	6-11	0.826	0.922	0.922	0.001	0.003

S5. Context attribution

Table S6 reports the complete L1 information-set analysis. Occupational history has the largest estimated single-context gain and the largest leave-one-context-out drop. Regional and residual-market histories remain complementary because the full information set achieves the highest AP.

Table S6 L1 context attribution under the strong-local specification. Effect is gain over strong local for single-context and full rows, and AP loss from full Mosaic for leave-one-context-out rows.

View	Information set	AP	Effect (ΔAP)
Baseline	Strong local	0.320	-
Single context	Regional only	0.340	0.020
Single context	Occupational only	0.357	0.037
Single context	Residual market only	0.339	0.019
Full	Full Mosaic	0.371	0.051
Leave one out	Full - regional	0.358	0.013
Leave one out	Full - occupational	0.340	0.031
Leave one out	Full - residual market	0.365	0.006

S6. Learner and static-identity controls

The logistic-linear control uses the same strong focal-memory representation as the primary analysis. The neural comparison uses a parameter-matched pair with common static identities. Both learner comparisons show a positive cross-scale gain (Table S7).

Table S7 Additional learner-class comparisons.

Learner	Information set	Rare-dormant AP
Logistic-linear	Strong local	0.309
Logistic-linear	Strong local + cross-scale	0.357
Secondary neural	Local histories + static IDs	0.241
Secondary neural	Cross-scale histories + static IDs	0.288

Adding skill, region and occupation identities to the L1 strong local representation retains a positive cross-scale gain (Table S8). This control tests whether the cross-scale advantage persists after static skill, region and occupation identities are included; it does not fully absorb pair-specific or time-varying propensities.

Table S8 Static-identity control under the strong focal-memory representation.

Level	Local specification	Local AP	Full Mosaic AP	Gain
L1	Strong local	0.320	0.371	0.051
L1	Strong local + static IDs	0.340	0.382	0.041

S7. Reproducibility

Online Resource 2 contains lattice construction, primary strong-local and full-Mosaic experiments, source attribution, learner checks, static-identity control, cluster bootstrap, source-data provenance and figure generation. The test suite checks lattice arithmetic, input integrity, context disjointness and event definitions. SHA256SUMS.txt verifies the pristine code archive, while build_frozen_results.py checks the cited compact result tables for presence and readability and regenerates Fig. 1 and Fig. 2 into a separate output directory.